



BAYESIAN ATTENTION MECHANISM

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A PROBABILISTIC FRAMEWORK
FOR POSITIONAL ENCODING AND
CONTEXT LENGTH EXTRAPOLATION

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Motivation

Transformer-based language models rely on positional encoding (PE) to handle **token order** and support **context length extrapolation**.

Existing PE methods **lack theoretical clarity** and rely on **limited evaluation metrics** to substantiate their extrapolation claims.

Bayesian Attention Mechanism

We propose the **Bayesian Attention Mechanism** (BAM), built on a simple reframing: self-attention as an **expectation of values** under a probability distribution over tokens.

$$\text{self-attention}(\mathbf{q}_i, \mathbf{K}, \mathbf{V}) = \frac{\exp(\text{score}(\mathbf{q}_i, \mathbf{k}_j))}{\sum_z \exp(\text{score}(\mathbf{q}_i, \mathbf{k}_z))} \mathbf{V} \\ = \sum_j p_{ij} \mathbf{v}_j = \mathbf{E}_{j|i}[\mathbf{V}]$$

The distribution is where positional information lives. Factoring it over content and position, **PE emerges as a prior**:

$$p_{ij} = \underbrace{p(f_{\text{cont}}(q_i, k_j))}_{\text{content likelihood}} \cdot \underbrace{p(g_{\text{pos}}(i, j))}_{\text{positional prior}}$$



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Additive Relative Positional Encodings

Theorem 1. Any additive $\text{score}(q_i, k_j) = f_{\text{cont}}(q_i, k_j) + g_{\text{pos}}(i, j)$ factors the attention weight into a **content likelihood** $\times$ **positional prior**:

$$p_{ij} = \frac{\exp(f_{\text{cont}}(\mathbf{q}_i, \mathbf{k}_j) + g_{\text{pos}}(i, j))}{\sum_z \exp(f_{\text{cont}}(\mathbf{q}_i, \mathbf{k}_z) + g_{\text{pos}}(i, z))} = \underbrace{p(f_{\text{cont}}(\mathbf{q}_i, \mathbf{k}_j))}_Z \cdot \underbrace{p(g_{\text{pos}}(i, j))}$$

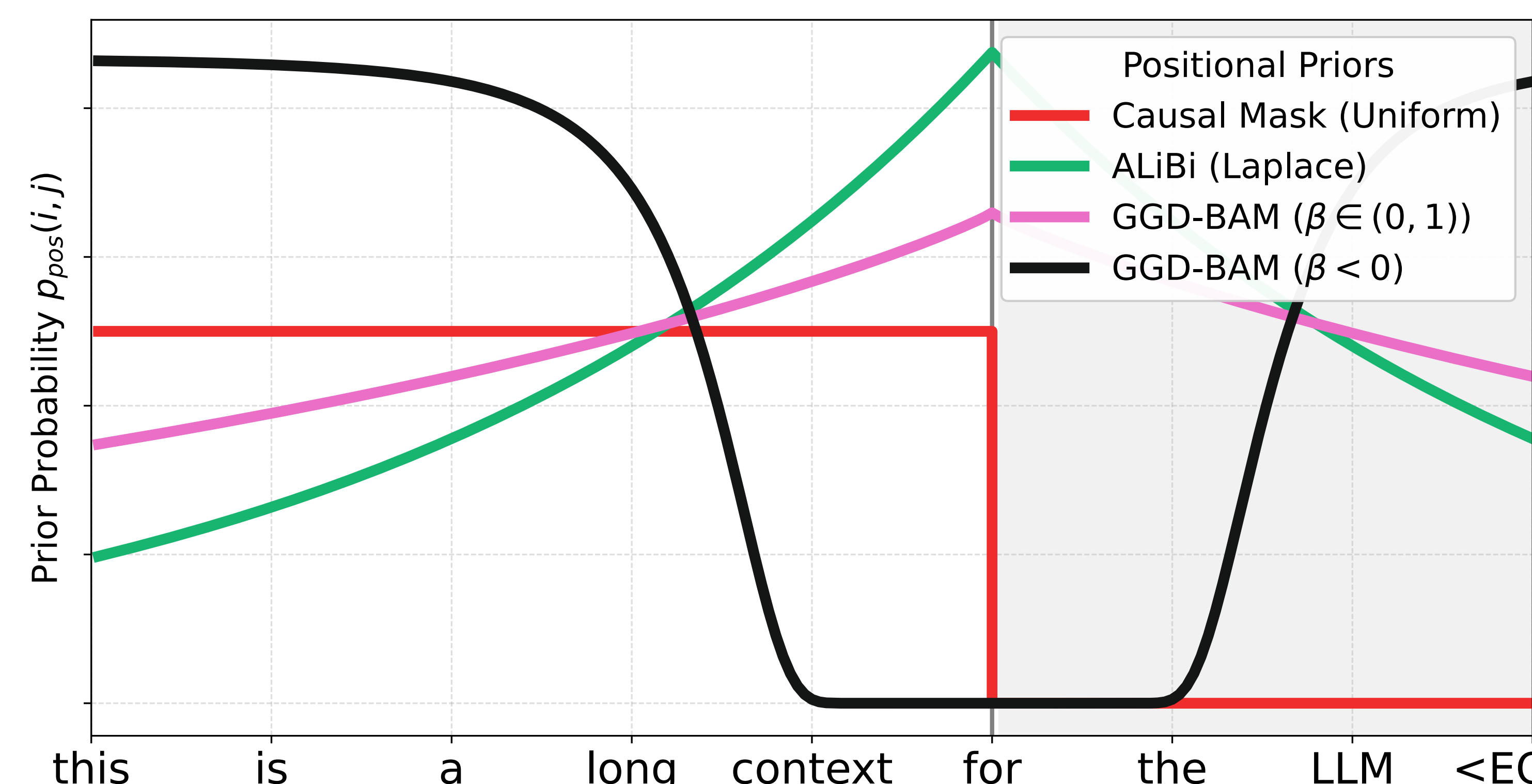
We instantiate BAM with the **Generalized Gaussian Distribution** (GGD), giving the following bias function:

$$g_{\text{pos}}(i, j) = - \left| \frac{j - i - \mu}{\alpha} \right|^\beta$$

GGD unifies the Laplace ($\beta = 1$), Normal ($\beta = 2$), and even heavier-tailed distributions ($0 < \beta < 1$) in a single family.

Since positions are discrete and bounded, the prior itself can represent unbounded distributions ($\beta < 0$).

This enables **BAM (GGD)** to attend to **arbitrarily long contexts**.



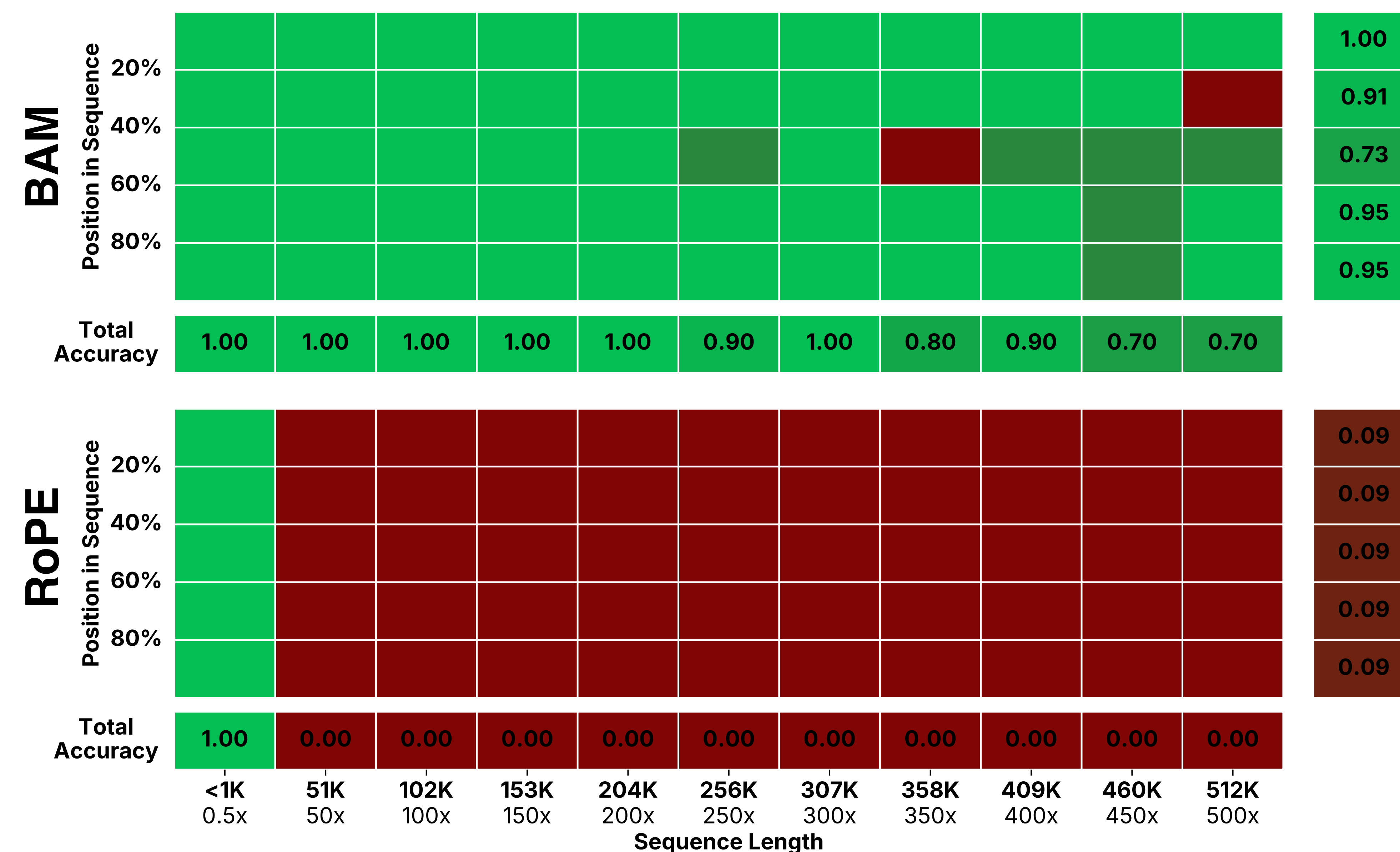
Matrix implementation of **BAM (GGD)**:

$$\mathbf{QK}^T + \mathbf{M} - \frac{1}{\alpha} \times \mathbf{B} - \mu$$

Extrapolation Results

Here we present the results for **1B parameter** models trained on sequence lengths of up to **1024 tokens**. (Passkey Retrieval)

*RoPE is evaluated with a SOTA context-extension method ($\sim 10 \times$ extrapolation)

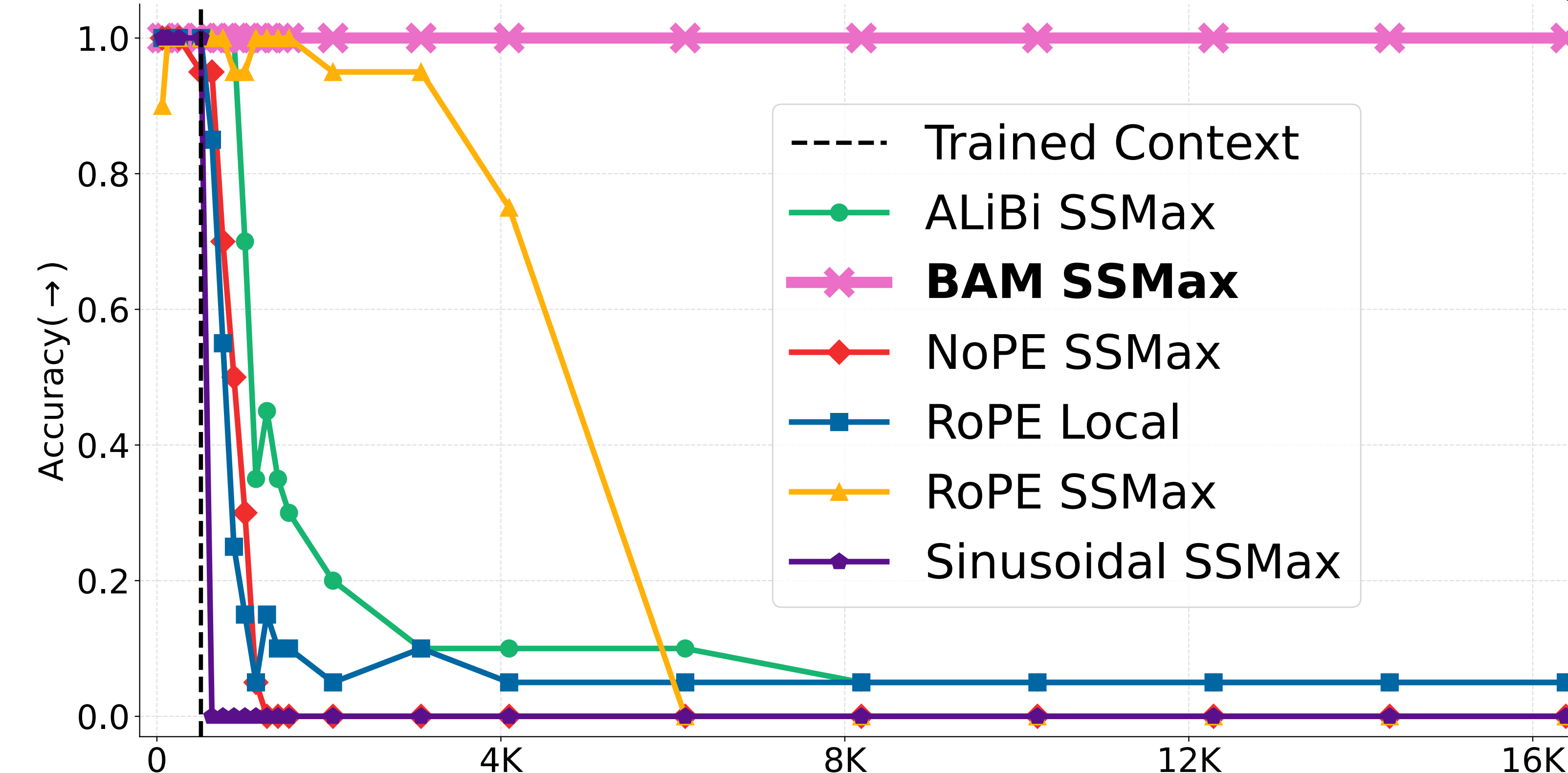


Accuracy heatmap evaluating performance across varying total sequence lengths (x-axis) and relative depth positions within the context (y-axis). Marginal blocks display the aggregated average accuracy per column and row.

Our PE achieves SOTA context length extrapolation in passkey retrieval; Retaining 70% accuracy even at 500 $\times$ the trained context length.

Passkey Retrieval:

- Trained on 512 Tokens
- 120M Parameters



Performance Results (1B Models)

	MMLU	ARC-Easy	ARC-Challenge	LongBenchv2
BAM (GGD)	0.3716	0.5770	0.4132	28.6
RoPE (extended)	0.3573	0.5715	0.4123	24.2